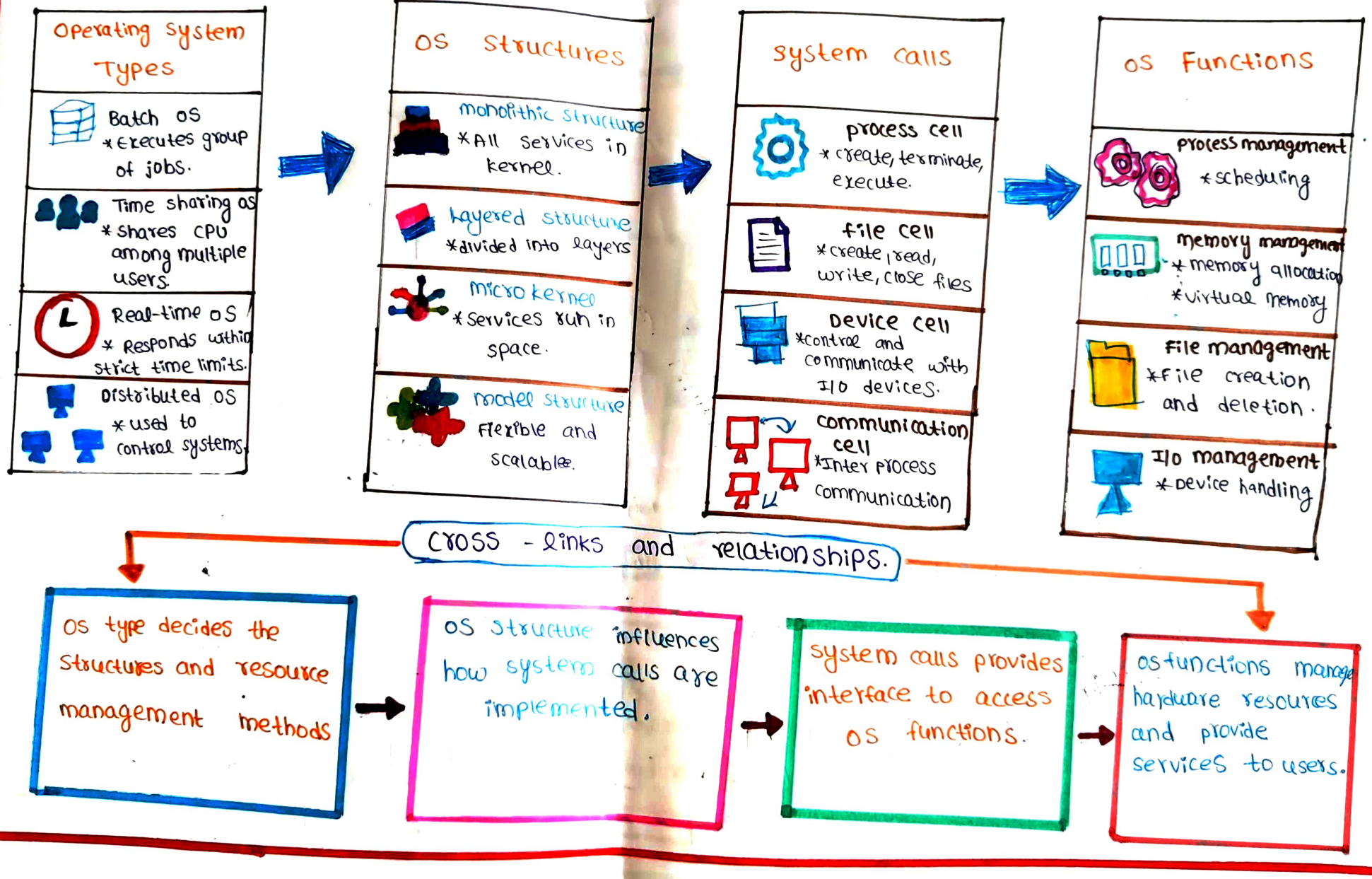
CO1- Assessment tool-2

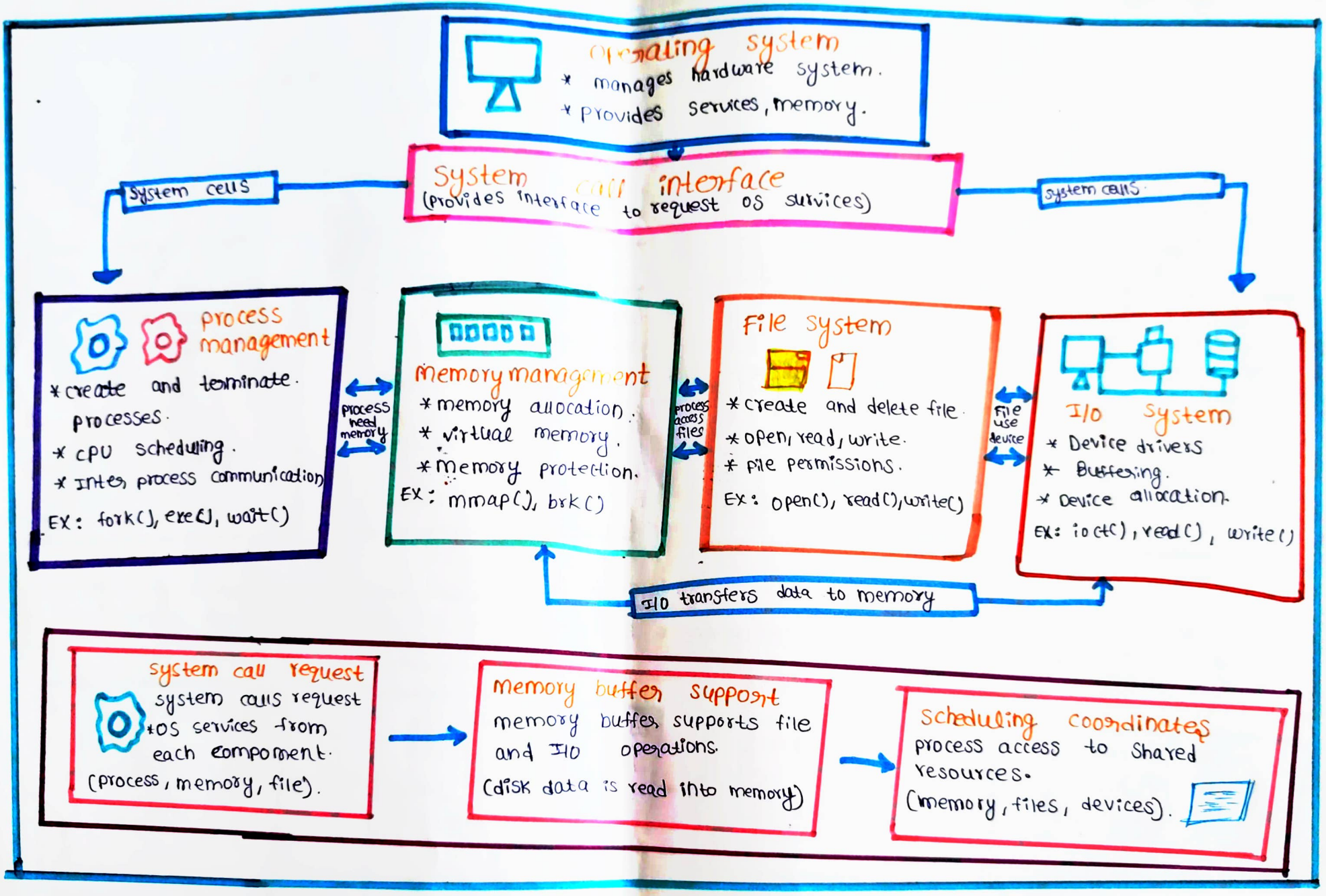
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1) Operating system Types → OS structures → system calls → OS functions.



Comprehensive concept map

2)



Comparative concept map

3)

Operating system Types

* Different OS types are designed for different needs.

Batch OS

* Executes grouped jobs.
* Little / no users.

Advantages

* High job throughput

Limitation

* Slow response time.

Time-sharing OS

* Sharing CPU among users
* Inactive operation.

* **Advantages**
  Quick response time.

Limitation

* Resource contection.

Distributed OS

* Share resource workload
* shares resources.

Advantages

* Resource sharing.

Limitation

* Network dependency.

Real-time OS

* Responds within deadlines.
* used in control system.

Advantages

* predictable timings.

Limitation.

* strict timing.

Similarities.

* All manage hardware resources.
* Execute programs
* manage processes & memory
* provide services to users.

Batch OS

Best for repeated large jobs.

Time sharing OS

Best for inactive users.

Distributed OS

Best for network. resource sharing.

Real-time OS.

Best for deadline critical tasks.

Dynamic System call Execution Flow

4)

Execution Flow

① User program

* calls a library function.

Ex: read(), write().

* uses API to request system services.

② API

* provides a simple interface to the user.
* converts function call into system call.

③ System call interface

* switches from user mode to kernel mode.
* Transfers to kernel.

④ Kernel

Check permissions (if valid or not).

* performs the requested operations.

⑤ Response to user

* kernel returns result or error codes.
* API returns result to user program.

if invalid

& invalid

Permission Check

* if not allowed → return error code.
* allowed → continue with operation.

interrupt

Resumes execution

Interrupt Handler

* Handles hardware interrupt.
* Reads completion status.

Process Scheduler

* Decides next process.
* may resume the current or switch to another process.

Examples system calls

Process	Files	Devices	Information
fork(), exec(), exit().	open(), read(), write()	ioct(), read(), write().	getpid(), time(), uname().

5)

User program

* needs OS services.
* calls system functions.

System calls

Process system calls

Ex: fork(), exec(), wait(), exit().

Related OS services

* process management.
* CPU scheduling.

Scheduling

(Allocates CPU to processes).

File system calls

Ex: open(), read(), write(), close().

Related OS services

* File management.
* store management

Memory allocation & storage access

manage file data in memory and disk.

Device System calls

Ex: ioctl(), read(), write(), open().

Related OS services

* Device handling.
* I/O management
* Interrupt handling

Device Handling

controls I/O devices and device drivers.

Information System calls

Ex: getpid(), time(). uname(), gettime().

Related OS services

* system information
* monitoring.
* Resource management.

Memory allocation & system information

provides system status and resource details.